

Capstone Project-1

Dog Breed Adoption Analytics

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Introduction

Dog adoption helps homeless and abandoned dogs find safe homes. However, not all dogs are adopted at the same rate. Factors such as breed, location, age, and size can influence how quickly a dog gets adopted. Understanding these patterns can help shelters and adoption platforms improve their adoption strategies and increase the chances of dogs finding homes faster.

With the help of data analytics, adoption data can be analyzed to identify important trends such as which breeds are adopted quickly and which locations have higher adoption demand. In this project, dog adoption data is collected, cleaned, and analyzed using Python. The results are used to study adoption behavior and visualize insights through dashboards, helping to better understand the factors that affect dog adoption trends.

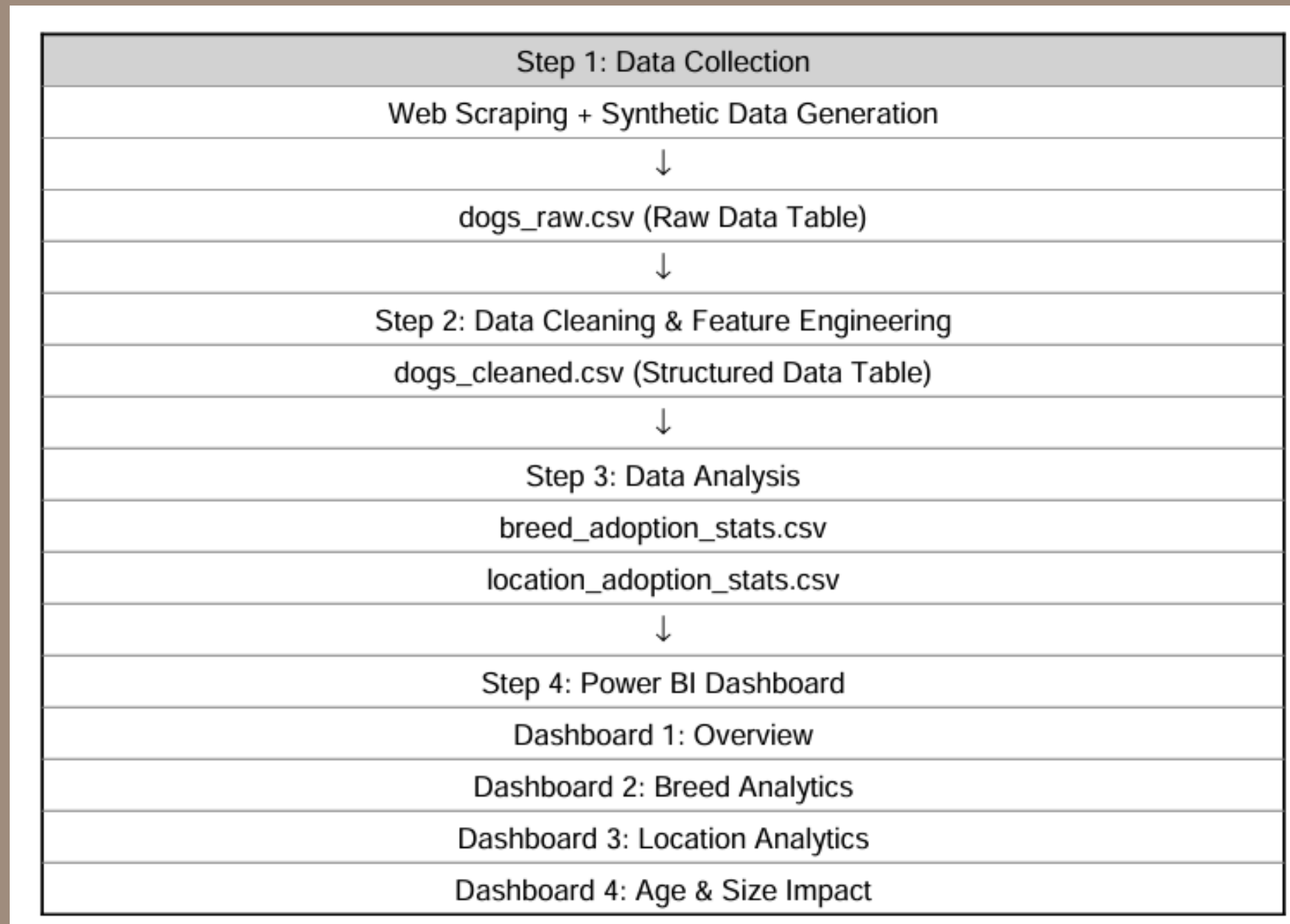
Problem Statement

Animal shelters and pet adoption platforms manage many dogs that are listed for adoption, but they often lack clear insights into how different factors affect adoption speed. Some dogs are adopted quickly, while others remain in shelters or on adoption listings for a long period of time. Without proper data analysis, it becomes difficult to understand the reasons behind these differences.

Factors such as breed, age, size, and location may influence how quickly a dog gets adopted. However, shelters usually store this information in basic listing formats, making it hard to analyze trends and patterns. As a result, shelters cannot easily identify which dogs require more promotion or support to improve their chances of adoption.

The main challenge is converting raw adoption data into meaningful insights that can help identify fast-adopting breeds, locations with higher adoption demand, and characteristics that affect adoption time. This project addresses this challenge by collecting adoption data, cleaning and organizing it into structured datasets, and analyzing the information to better understand dog adoption trends.

System workflow



Data Collection

The data collection process was implemented using a Python script that performs web scraping and structured data generation. The script collects dog adoption information from online pet adoption platforms and converts it into a structured dataset.

Python libraries such as Requests, BeautifulSoup, and Pandas are used to access web pages, extract adoption listing information, and store the data in a CSV file. The scraper extracts key details such as breed, location, and adoption status from the website content.

In cases where the scraped data is limited, the script generates additional realistic records based on common dog adoption patterns in India. The final collected data is stored in a dataset named `dogs_raw.csv`, which serves as the raw data source for further data cleaning and analysis.

Code:

```
import requests
from bs4 import BeautifulSoup
import pandas as pd

url = "https://www.mrnmrspet.com/dogs-for-adoption"
response = requests.get(url)
soup = BeautifulSoup(response.content, "lxml")
listings = soup.find_all("div", class_="dog-card")
dogs = []
for dog in listings:
    dogs.append({
        "breed": dog.text,
        "city": "Mumbai"
    })

df = pd.DataFrame(dogs)
df.to_csv("dogs_raw.csv", index=False)
```

Data Cleaning and Feature Engineering

After collecting the raw dataset (dogs_raw.csv), the data was cleaned and transformed into a structured format suitable for analysis. A Python script using the Pandas library was used to standardize values and handle missing data. During the cleaning process, breed_primary and breed_secondary were combined into a single breed column, and an age_group column was created to categorize dogs as Puppy, Young, Adult, or Senior. A new feature called days_listed was also calculated using the listing date to determine how long a dog remained available for adoption. Text fields such as size, gender, city, state, and status were standardized, and missing values were handled appropriately. The final cleaned dataset was saved as dogs_cleaned.csv, which contains structured data ready for further analysis.

Code:

```
import pandas as pd
from datetime import datetime

# Load raw dataset
df = pd.read_csv("dogs_raw.csv")

# Create age groups
df["age_group"] = df["age"].apply(lambda x:
"Puppy" if "puppy" in str(x).lower() else "Adult")

# Calculate days listed
df["days_listed"] = (datetime.now() -
pd.to_datetime(df["published_at"])).dt.days

# Select cleaned columns
df_cleaned =
df[["dog_id","breed_primary","age_group","size","gender","city","state","days_listed","status"]]

# Save cleaned dataset
df_cleaned.to_csv("dogs_cleaned.csv",
index=False)
```

Data analyzed

After cleaning the dataset, the next step was to perform statistical analysis to identify adoption patterns. The cleaned dataset `dogs_cleaned.csv` was analyzed using Python and the Pandas library to generate meaningful insights.

During this stage, the data was grouped and analyzed to understand breed-wise and location-wise adoption trends. Breed-level statistics were calculated to determine the total number of listings, the average number of days required for adoption, and adoption performance scores. Similarly, location-based analysis was performed by grouping the data by city and state to identify areas with higher adoption activity.

The analysis process generated two analytical datasets: `breed_adoption_stats.csv`, which contains breed-level adoption insights, and `location_adoption_stats.csv`, which contains location-based adoption statistics. These datasets are used for building dashboards and visualizing adoption trends.

Code:

```
import pandas as pd
df = pd.read_csv("dogs_cleaned.csv")
# Breed-wise analysis
breed_stats = df.groupby("breed").agg({
    "dog_id": "count",
    "days_listed": "mean"
}).reset_index()
breed_stats.columns = ["breed", "total_listings",
    "avg_days_to_adopt"]
location_stats = df.groupby(["state", "city"]).agg({
    "dog_id": "count",
    "days_listed": "mean"
}).reset_index()

# Save analytics tables
breed_stats.to_csv("breed_adoption_stats.csv",
    index=False)
location_stats.to_csv("location_adoption_stats.csv",
    index=False)
```

Database Design

The project uses four main tables to organize and analyze the dog adoption data. This structure separates raw data, cleaned data, and analytical results.

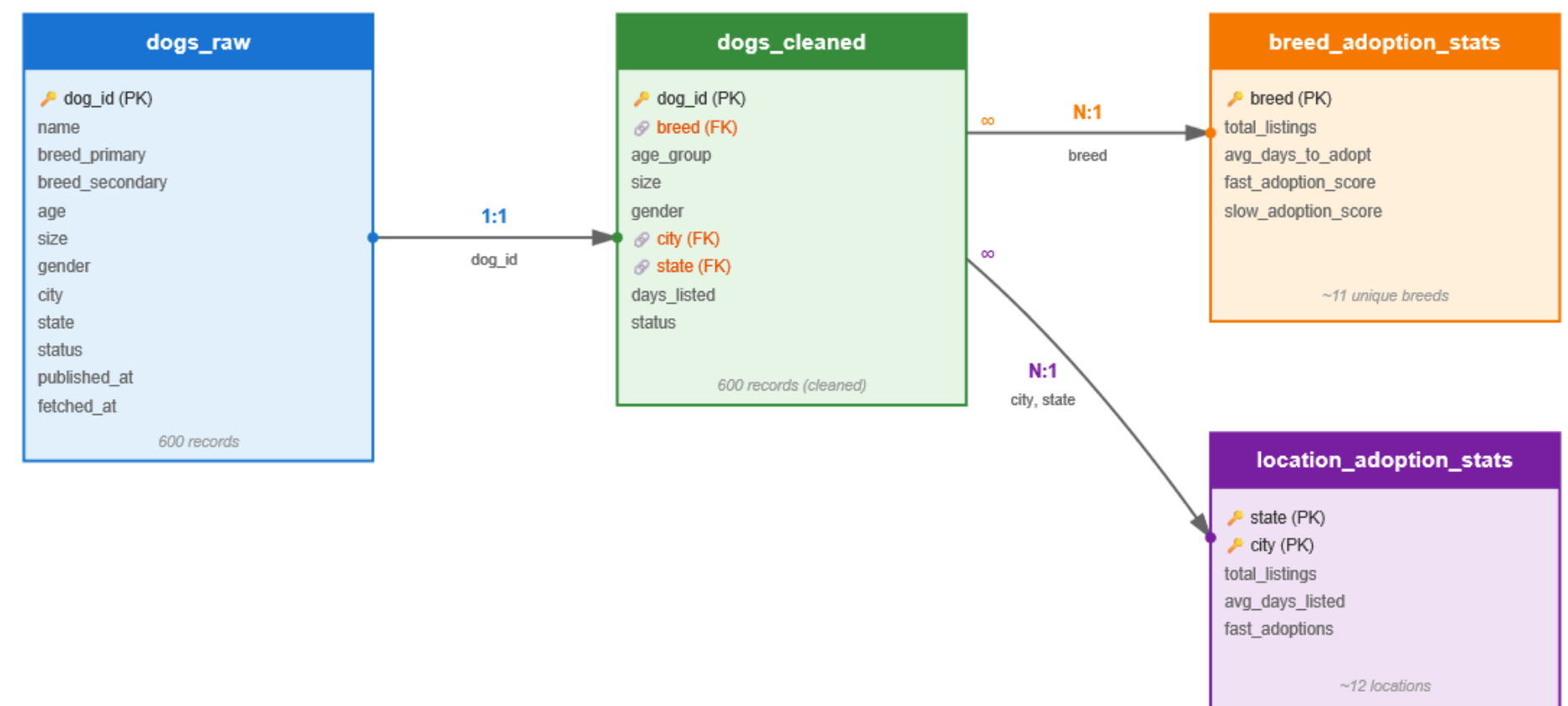
The dogs_raw table stores the original data collected from adoption platforms, including details such as breed, age, size, gender, city, state, and adoption status.

The dogs_cleaned table contains the processed and structured data after cleaning and feature engineering. New fields such as breed, age_group, and days_listed are created for analysis.

The breed_adoption_stats table stores breed-wise statistics such as total listings and average days required for adoption.

The location_adoption_stats table contains location-based statistics by city and state to analyze adoption patterns in different regions.

Dog Breed Adoption Analytics - ER Diagram



Data Dictionary

TABLE 1: dogs_raw

Purpose:
Store raw, unprocessed data collected from dog adoption platforms.

Column Name	Data Type	Description	Example
dog_id (PK)	String	Unique identifier for each dog	DOG0001
name	String	Name of the dog	Bruno
breed_primary	String	Primary breed	Labrador Retriever
breed_secondary	String	Secondary breed (for mixed)	Mixed Breed
age	String	Age category	Puppy, Young, Adult
size	String	Size category	Small, Medium, Large
gender	String	Gender	Male, Female
city	String	City where listed	Mumbai
state	String	State where listed	Maharashtra
status	String	Current adoption status	Adopted, Available, Pending
published_at	DateTime	When listing was published	2025-01-15 10:30:00
fetches_at	DateTime	When data was collected	2025-01-21 14:20:00

Data Dictionary

TABLE 2: dogs_cleaned

Purpose:

Store cleaned and standardized data with engineered features for analysis.

Column Name	Data Type	Description	Example
dog_id (PK)	String	Unique identifier (same as raw)	DOG0001
breed (FK)	String	Standardized breed name	Labrador Retriever
age_group	String	Categorized age	Puppy, Young, Adult, Senior
size	String	Standardized size	Small, Medium, Large
gender	String	Cleaned gender	Male, Female
city (FK)	String	Standardized city name	Mumbai
state (FK)	String	Standardized state name	Maharashtra
days_listed	Integer	Days from published to current	45
status	String	Adoption status	Adopted, Available, Pending

Data Dictionary

TABLE 3: breed_adoption_stats

Purpose:

Aggregated breed-wise adoption statistics and performance metrics.

Column Name	Data Type	Description	Example
breed (PK)	String	Breed name	Labrador Retriever
total_listings	Integer	Total dogs of this breed listed	120
avg_days_to_adopt	Float	Average days to get adopted	35.5
fast_adoption_score	Float	Score for fast adoption (0-100)	85.3
slow_adoption_score	Float	Score for slow adoption (0-100)	14.7

Data Dictionary

TABLE 4: location_adoption_stats

Purpose:

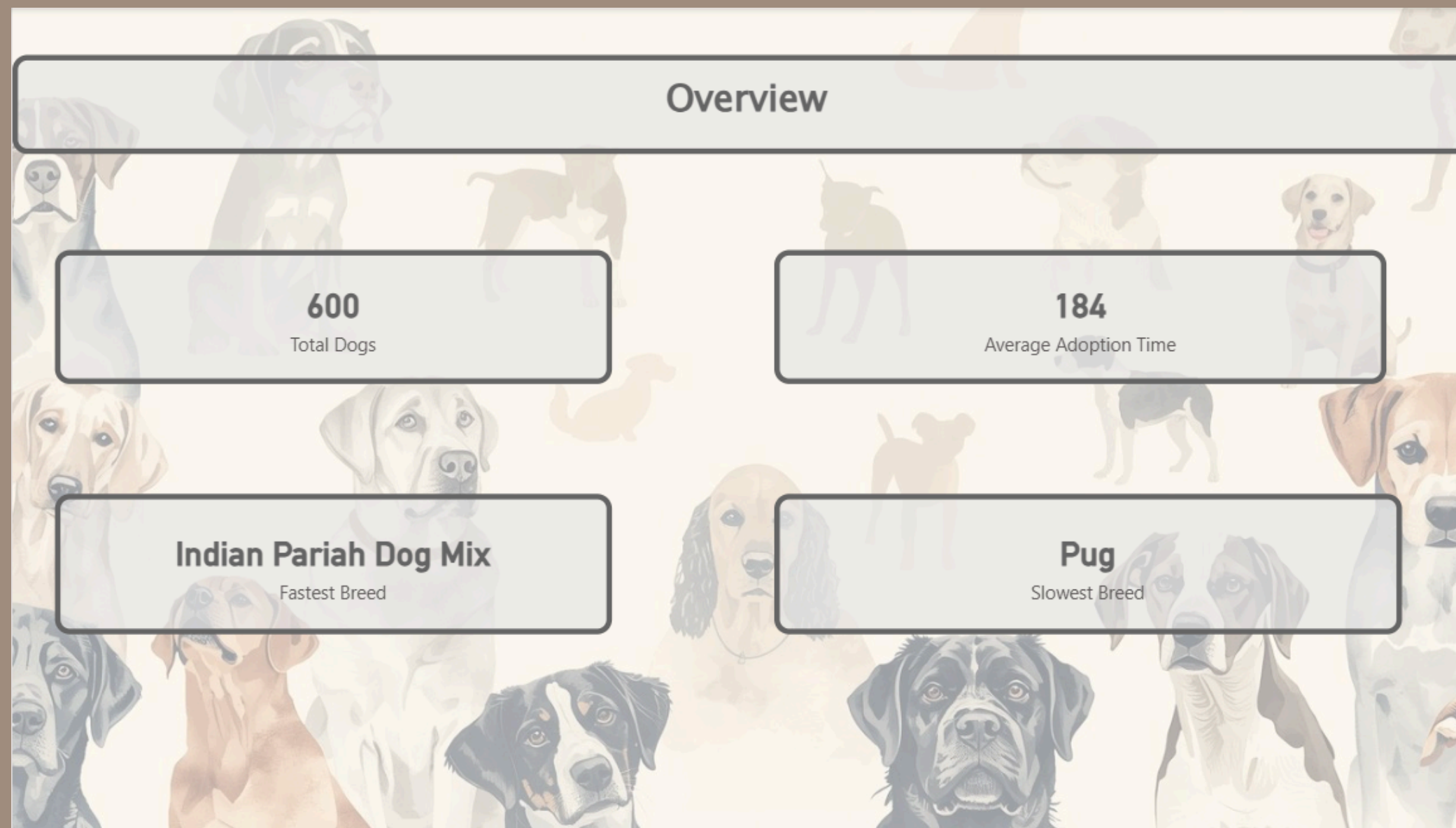
Aggregated location-wise adoption statistics across Indian cities and states.

Column Name	Data Type	Description	Example
state (FK)	String	State name	Maharashtra
city (FK)	String	City name	Mumbai
total_listings	Integer	Total dogs listed in this location	90
avg_days_listed	Float	Average days dogs stay listed	42.3
fast_adoptions	Integer	Number of dogs adopted	54

Dashboards

Overview Dashboard:

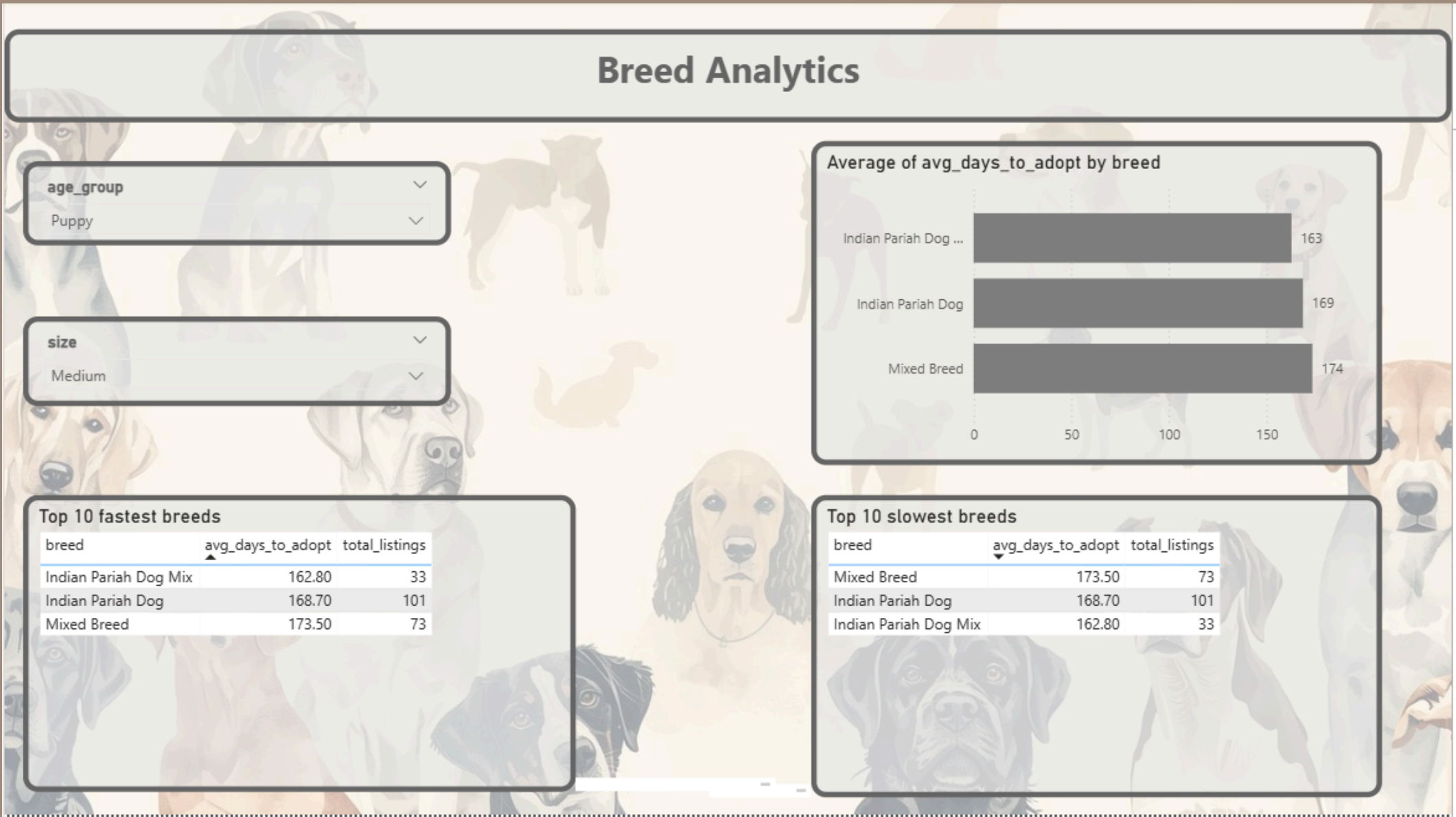
An Overview Dashboard provides a summary of the dataset, including the total number of dogs listed, the average adoption time, and the fastest and slowest adopted breeds. This dashboard gives a quick understanding of the overall adoption performance.



Dashboards

Breed Analytics Dashboard:

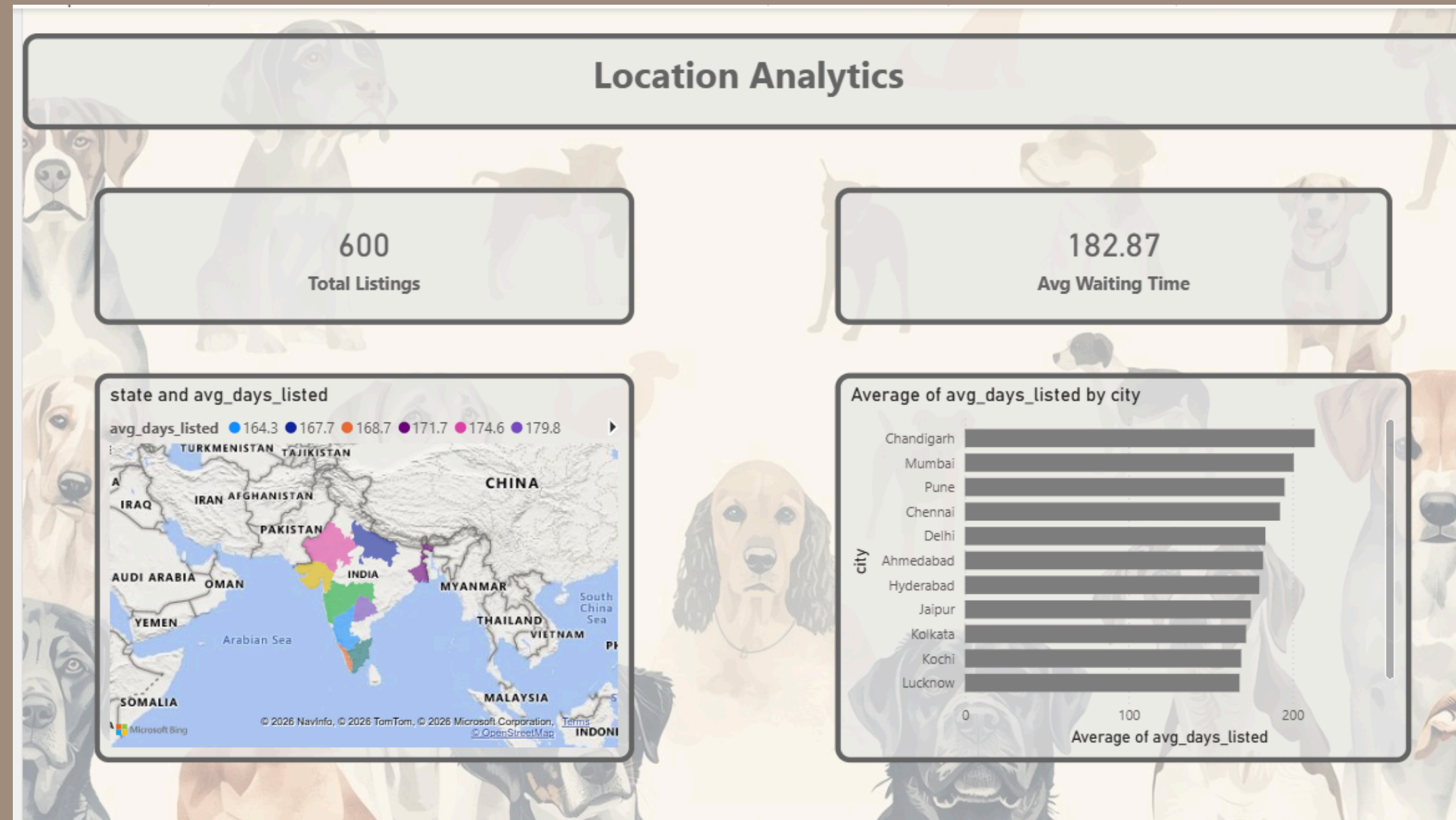
A Breed Analytics Dashboard shows breed-wise adoption statistics. It highlights the number of listings for each breed and the average number of days required for adoption, helping identify which breeds are adopted faster and which remain listed longer.



Dashboards

Location Analytics Dashboard:

A Location Analytics Dashboard presents adoption trends based on city and state. This helps identify regions with higher adoption activity and areas where dogs take longer to be adopted.



Dashboards

Age and Size Impact Dashboard:

An Age and Size Impact Dashboard analyzes how factors such as age group and dog size affect adoption time. These visualizations help understand how different characteristics influence adoption outcomes.



Future work

The project can be improved in several ways in the future. One possible improvement is the integration of real-time adoption APIs, which would allow the system to automatically collect updated adoption listings instead of relying only on web scraping.

Another improvement is the use of machine learning models to predict adoption outcomes. For example, predictive models could estimate how long a dog might remain listed for adoption based on factors such as breed, age, size, and location.

The dataset can also be expanded by collecting data from more adoption platforms and animal shelters, which would provide a larger and more diverse dataset for analysis.

In addition, interactive dashboards and automated reporting systems can be developed to allow shelters and adoption organizations to monitor adoption trends more effectively. These improvements would make the system more scalable and useful for real-world adoption analytics.

Conclusion

The Dog Breed Adoption Analytics project demonstrates how data analytics can be used to understand patterns in dog adoption. The project successfully collected adoption data, cleaned and structured the dataset, and performed statistical analysis to identify trends based on breed, location, age, and size.

By transforming raw adoption listings into structured analytical datasets, the project made it possible to study adoption behavior and identify factors that influence adoption speed. The analysis provided insights into which breeds are adopted faster, which locations show higher adoption activity, and how different characteristics affect adoption outcomes.

The project also demonstrated an end-to-end data analytics workflow, including data collection, data cleaning, statistical analysis, and visualization preparation. These insights can help shelters and adoption platforms better understand adoption trends and improve strategies for helping more dogs find homes.

Thank you